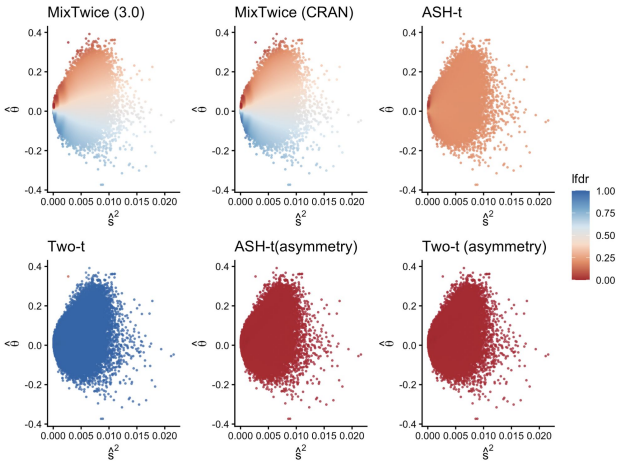
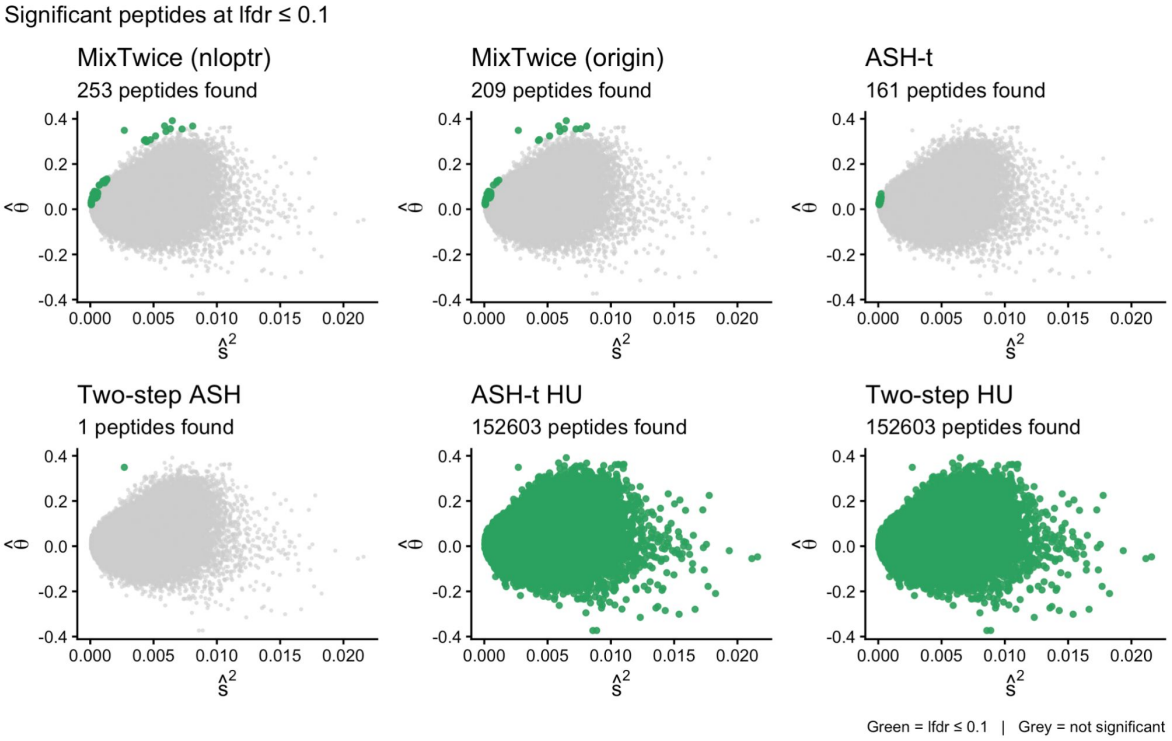
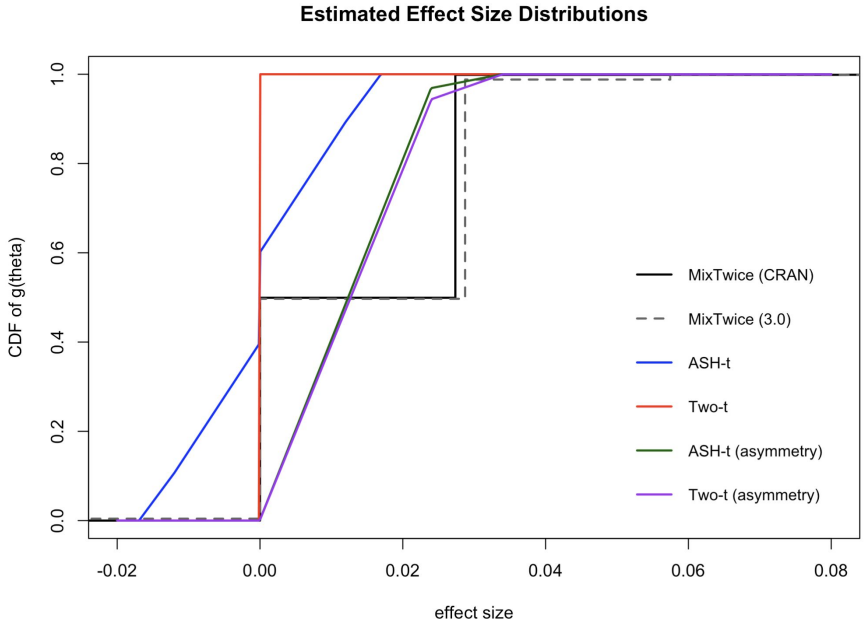
Weak Signal Data

```
> model <- MixTwice::mixtwice(
+   thetaHat = thetaHat,
+   s2       = s2,
+   Btheta   = 15,
+   Bsigma2  = 10,
+   df       = df,
+   prop     = 1,
+   "AugLag"
+ )
> sum(model$lfdr <= 0.1)
[1] 228
```





```
> model$grid.theta
 [1] -0.40550753 -0.37847369 -0.35143986
 [4] -0.32440602 -0.29737219 -0.27033835
 [7] -0.24330452 -0.21627068 -0.18923685
[10] -0.16220301 -0.13516918 -0.10813534
[13] -0.08110151 -0.05406767 -0.02703384
[16]  0.00000000  0.02703384  0.05406767
[19]  0.08110151  0.10813534  0.13516918
[22]  0.16220301  0.18923685  0.21627068
[25]  0.24330452  0.27033835  0.29737219
[28]  0.32440602  0.35143986  0.37847369
[31]  0.40550753
> model$mix.theta
 [1] 0.000000e+00 3.624513e-11 0.000000e+00
 [4] 1.105228e-10 2.226656e-11 0.000000e+00
 [7] 0.000000e+00 4.107512e-11 2.046210e-11
[10] 1.469132e-10 3.326178e-11 9.810797e-12
[13] 1.208401e-11 4.419888e-11 3.191119e-09
[16] 4.993345e-01 4.993345e-01 1.210065e-04
[19] 1.210002e-04 1.209949e-04 1.209946e-04
[22] 1.209944e-04 1.209943e-04 1.209941e-04
[25] 1.209939e-04 1.209860e-04 1.209842e-04
[28] 1.209805e-04 7.830083e-09 0.000000e+00
[31] 0.000000e+00
```



ASH-t: $g(\theta) = 0.2027\,\delta_0 + 0.0656\,U[-0.01198,\ 0.01198] + 0.7317\,U[-0.01694,\ 0.01694]$

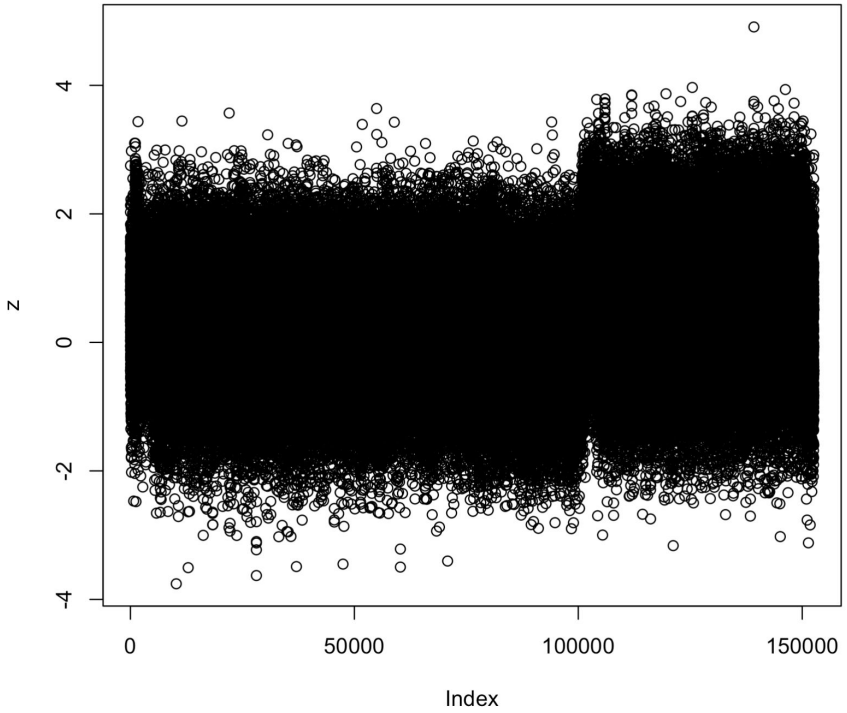
Two-step ASH-t: $g(\theta) = 0.9999556\,\delta_0 + 0.00004445\,U[-0.3844,\ 0.3844]$

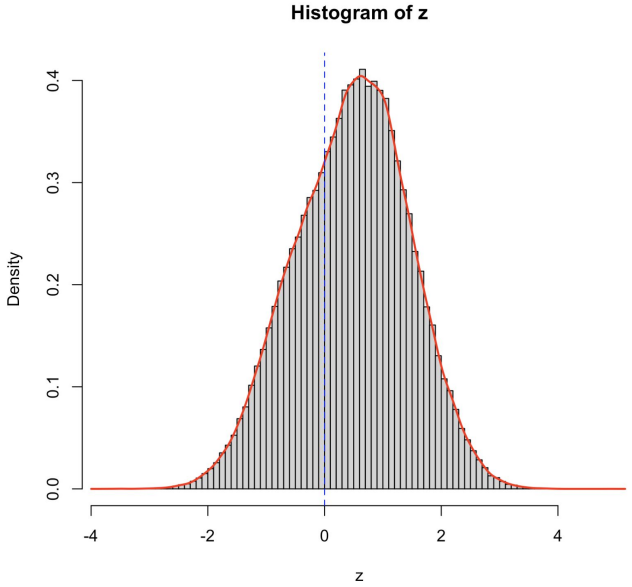
ASH-t (asymmetric): $g(\theta) = 0.00195\,\delta_0 + 0.8919\,U[0,\ 0.02396] + 0.1061\,U[0,\ 0.03389]$

Two-step ASH-t (asymmetric): $g(\theta) = 0.0020\,\delta_0 + 0.8072\,U[0,\ 0.0240] + 0.1907\,U[0,\ 0.0340] + 0.0001\,U[0,\ 0.3844]$

What happens with ASH (asymmetry)

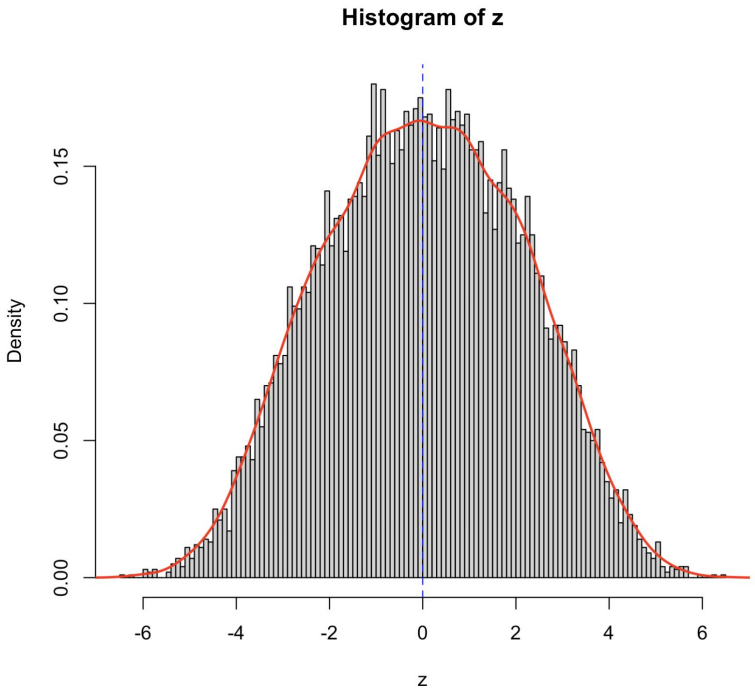
The observed data suggest that a single common prior g(θ) may be inadequate to describe the underlying effects due to the presence of two groups of data with different center.

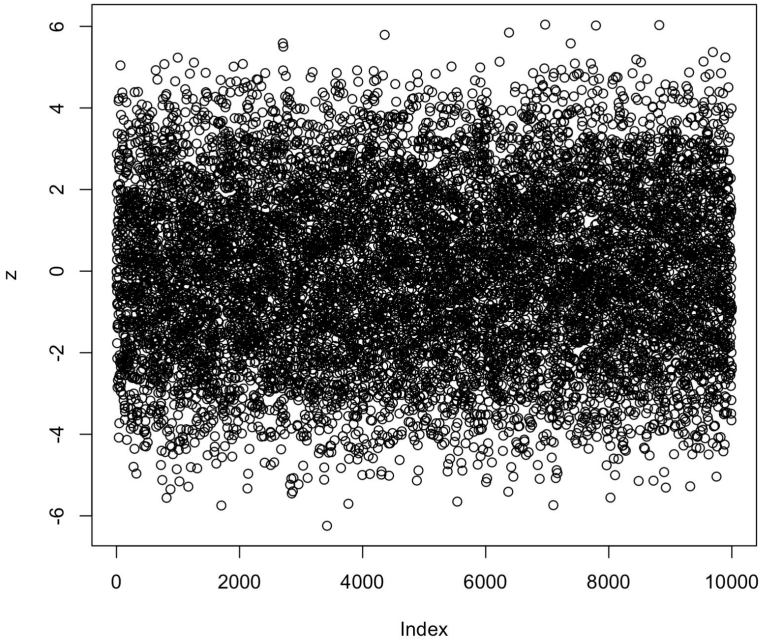




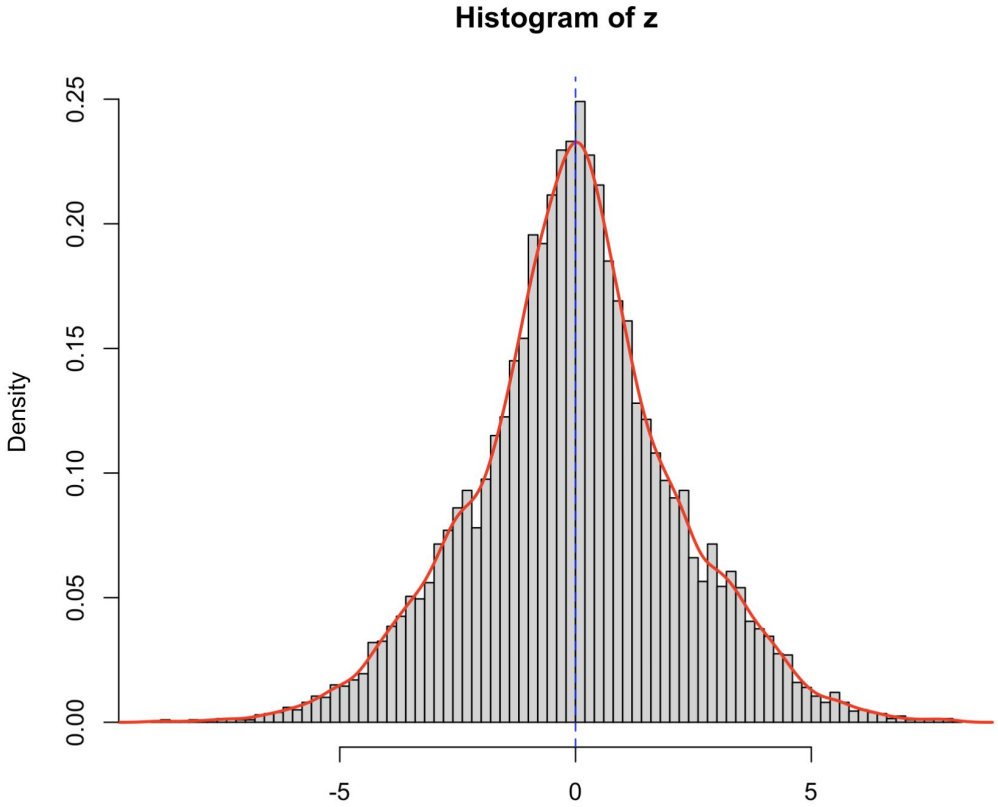
What happens with ASH (asymmetry)

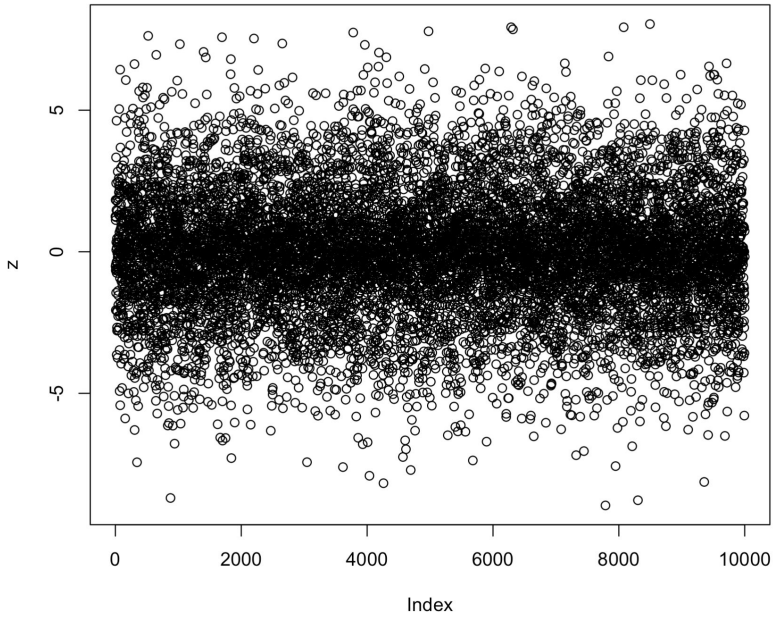
Simulations: (dependent)





Simulation (independent/bimodal)





Independent Simulation

$$\theta_i \sim \pi_0 \delta_0 + (1-\pi_0) g_{\text{alt}},$$

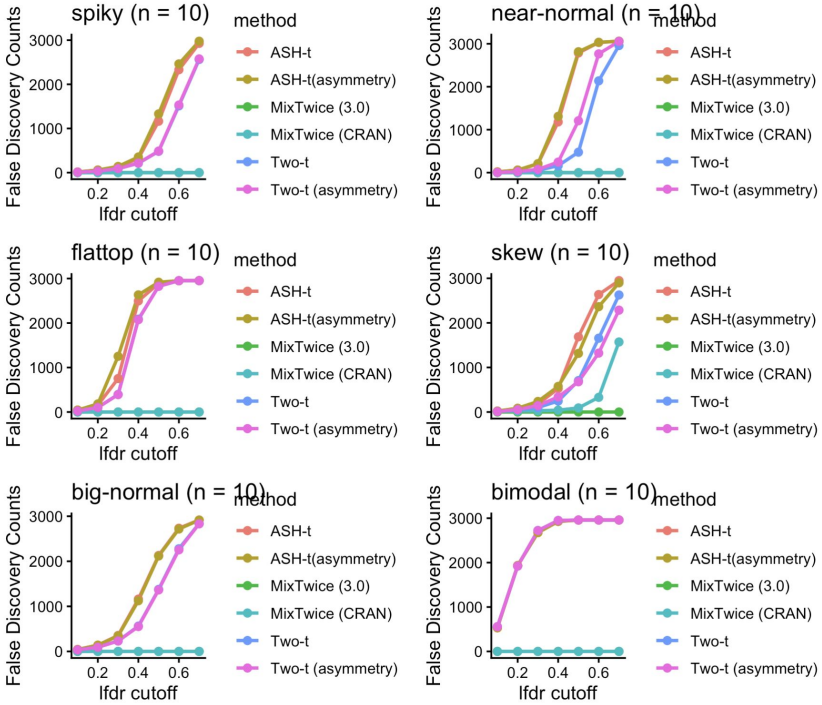
$$\sigma_i^2 \sim \text{Inverse-Gamma}(\alpha, \beta),$$

$$x_i \mid \theta_i, \sigma_i^2 \sim N(\theta_i, \sigma_i^2),$$

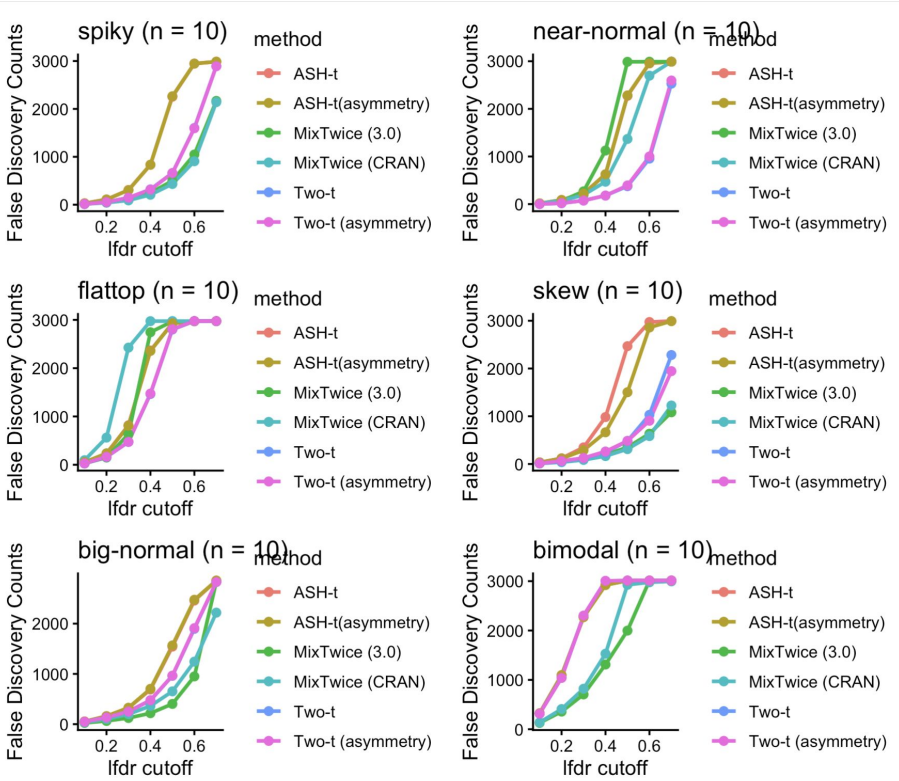
$$s_i^2 \mid \sigma_i^2 \sim \frac{\sigma_i^2}{\nu} \chi_\nu^2, \qquad \nu = 2n-2.$$

Scenario	g_{alt}
spiky	$0.2\,N(0, 0.3^2) + 0.3\,N(0, 0.8^2) + 0.3\,N(0, 1.5^2) + 0.2\,N(0, 3^2)$
near-normal	$0.55\,N(0, 0.5^2) + 0.3\,N(0, 0.9^2) + 0.15\,N(0, 1.4^2)$
flattop	$\frac{1}{7}\left[N(-1.2, 0.3^2) + N(-0.8, 0.45^2) + N(-0.4, 0.6^2) + N(0, 0.7^2) + N(0.4, 0.6^2) + N(0.8, 0.45^2) + N(1.2, 0.3^2)\right]$
skew	$0.3\,N(0, 0.4^2) + 0.3\,N(0.8, 0.8^2) + 0.25\,N(1.8, 1.4^2) + 0.15\,N(3.2, 2.2^2)$
big-normal	$0.35\,N(0, 1^2) + 0.4\,N(0, 2.2^2) + 0.25\,N(0, 3.5^2)$
bimodal	$0.25\,N(-2.5, 0.5^2) + 0.25\,N(-2.5, 1.5^2) + 0.25\,N(2.5, 0.5^2) + 0.25\,N(2.5, 1.5^2)$

Pi_0 = 0.3; n=10, “peptide” = 10000

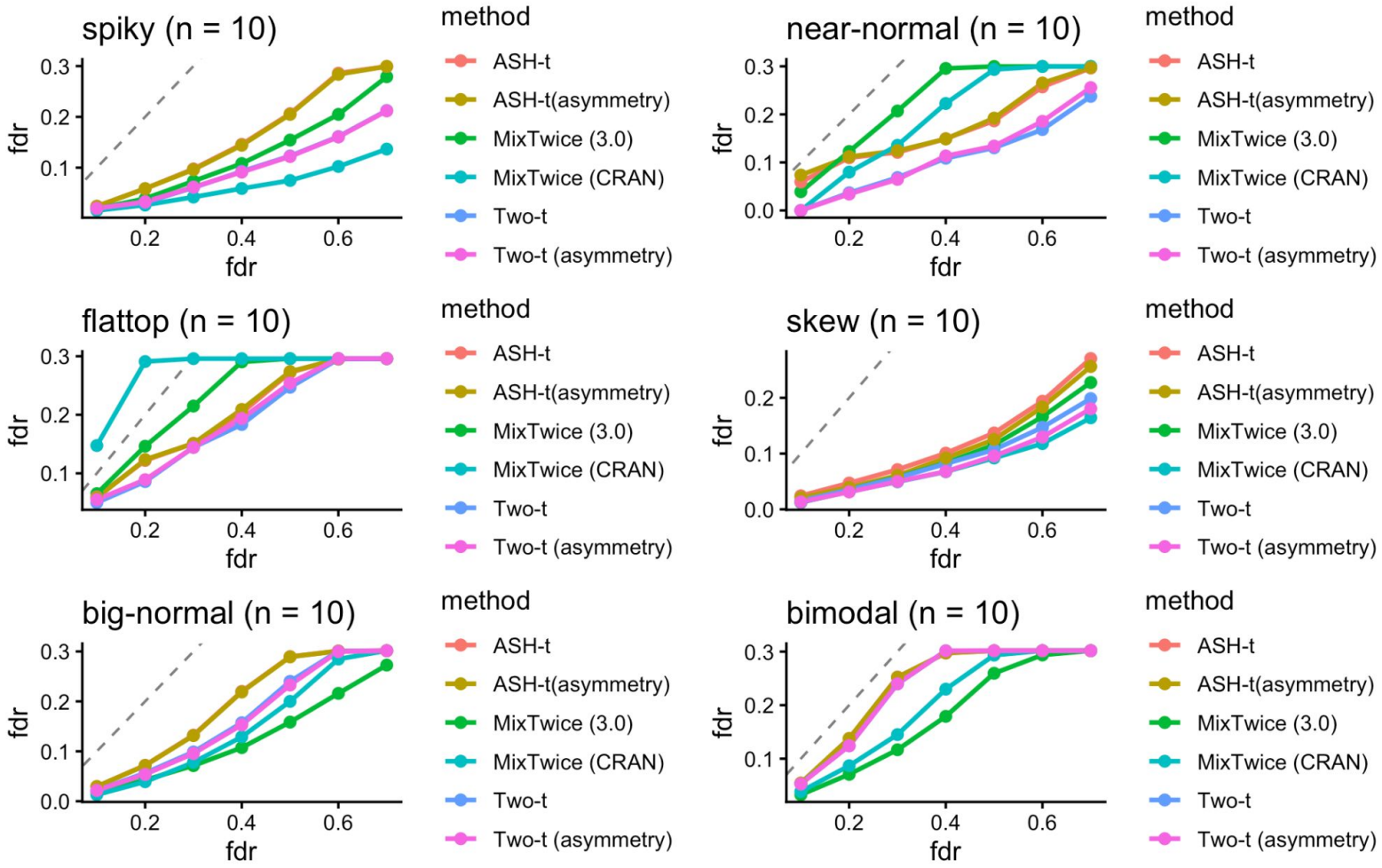


inv-gamma(1,1)

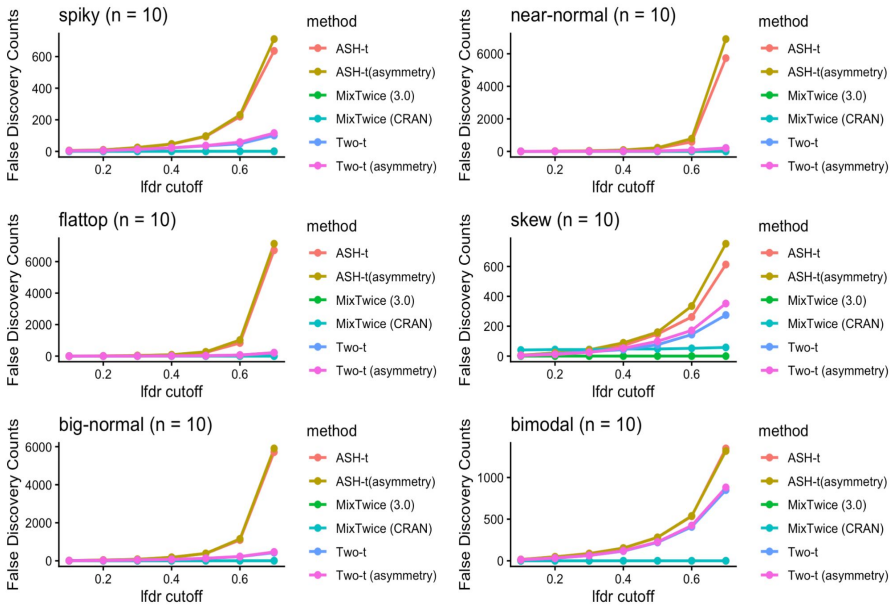


inv-gamma(3,2)

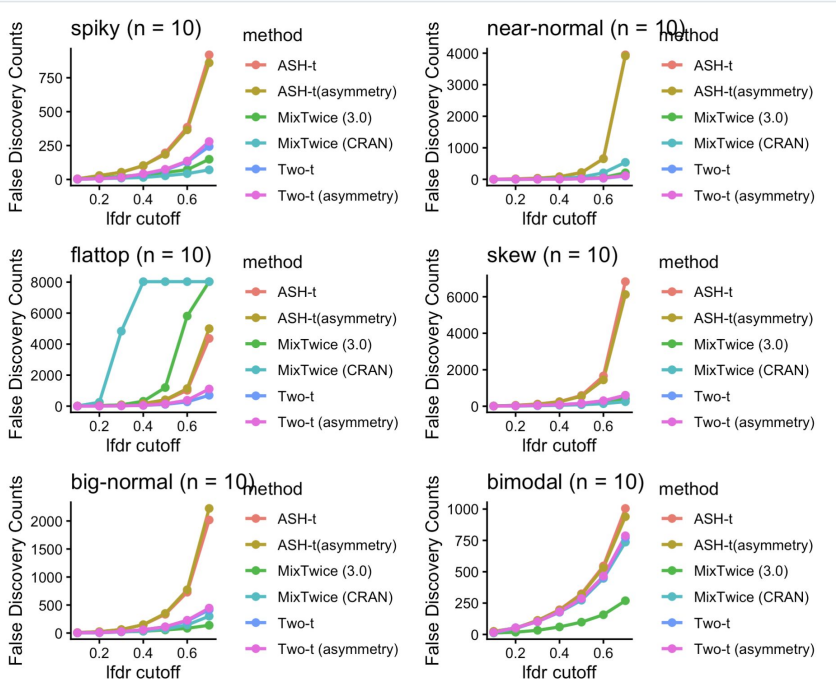
P


spiky (n = 10)
near-normal (n = 10)
flattop (n = 10)
skew (n = 10)
big-normal (n = 10)
bimodal (n = 10)
fdr
method
ASH-t
ASH-t(asymmetry)
MixTwice (3.0)
MixTwice (CRAN)
Two-t
Two-t (asymmetry)

Pi_0 = 0.8; n=10,

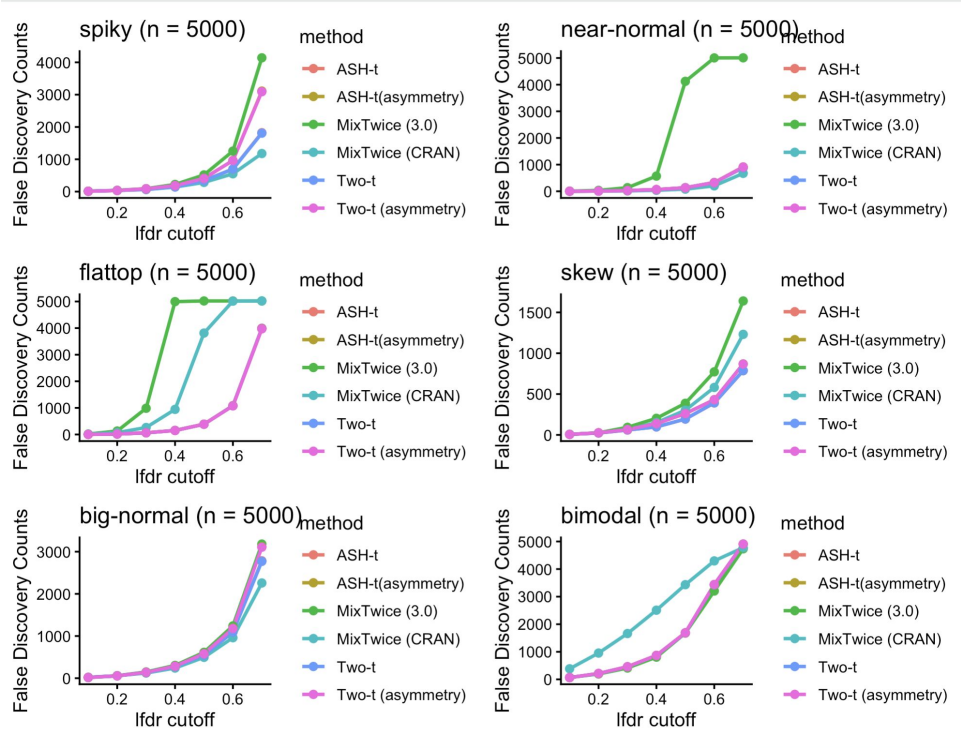


inv-gamma(1,1)



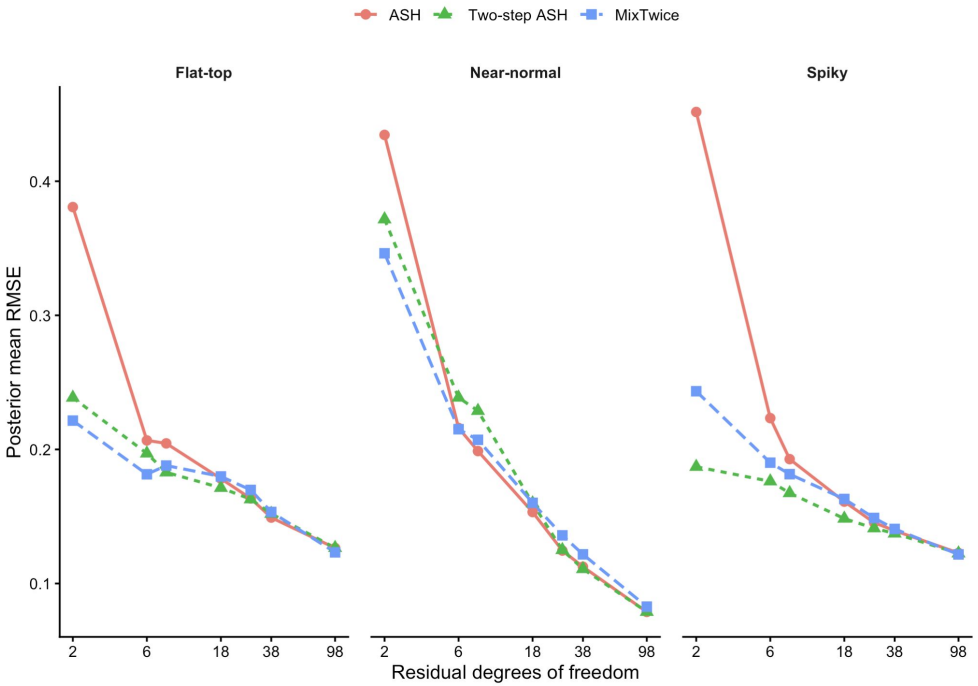
inv-gamma(3,2)

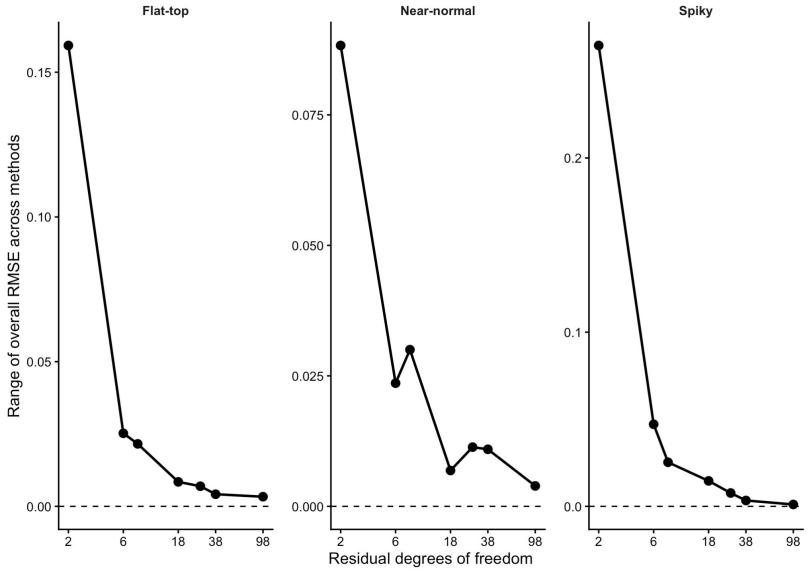
The check: pi_0= 0.5, n=5000



inv-gamma(5,5)

Check (3.3 Validation Under Diminishing Variance Uncertainty)





Dependent simulation

$$\sigma_i^2 \sim h(\sigma^2),$$

$$\sigma_i^2 \sim 0.88 \cdot 0.11\,\mathrm{Beta}(0.85, 6.5) \;+\; 0.12 \cdot 0.11\,\mathrm{Beta}(2.0, 4.2)$$

$$\gamma_i \sim \mathrm{Bernoulli}(1 - \pi_0),$$

$$\gamma_i \sim \mathrm{Bernoulli}(0.7)$$

$$\theta_i \mid \sigma_i^2, \gamma_i \sim \begin{cases} 0, & \gamma_i = 0, \\ g_{\mathrm{alt}}(\theta \mid \sigma_i^2), & \gamma_i = 1, \end{cases}$$

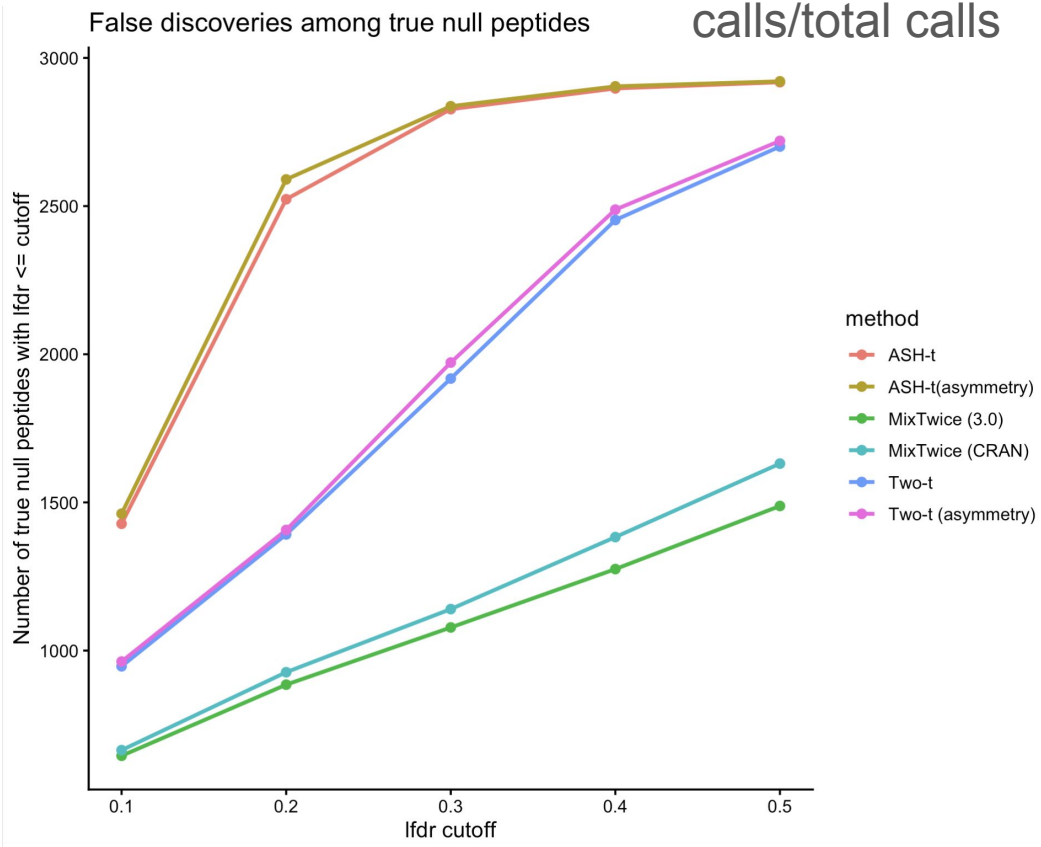
$$\theta_i \mid \sigma_i^2, \gamma_i \sim \begin{cases} 0 & \gamma_i = 0 \\ \mathcal{N}(0,\ 0.5\,\sigma_i^2) & \gamma_i = 1 \end{cases}$$

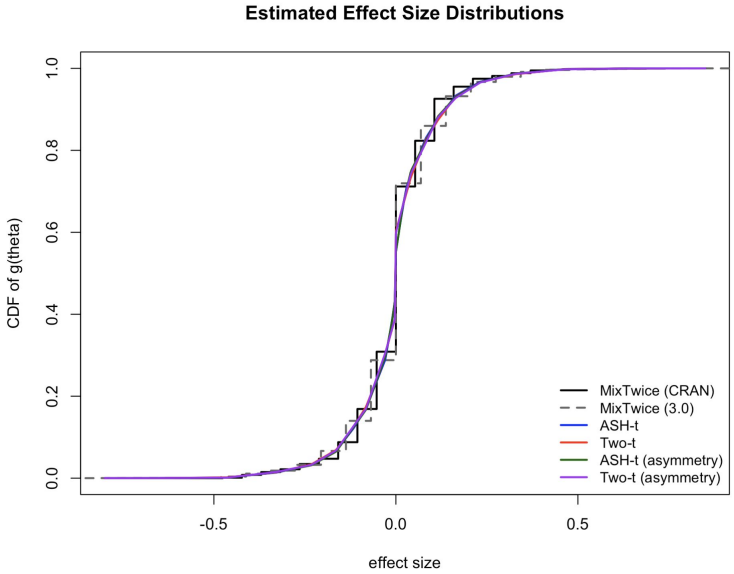
$$x_i \mid \theta_i, \sigma_i^2 \sim N(\theta_i, \sigma_i^2)$$

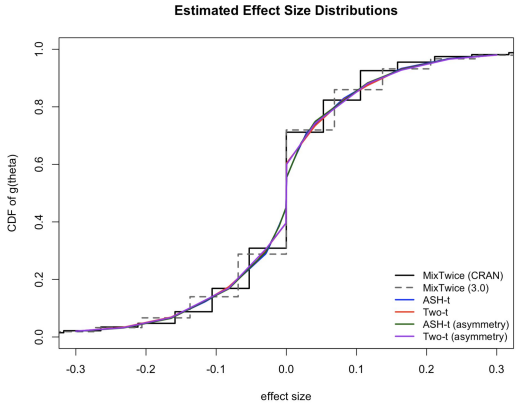
$$s_i^2 \mid \sigma_i^2 \sim \frac{\sigma_i^2}{\nu} \chi_\nu^2.$$

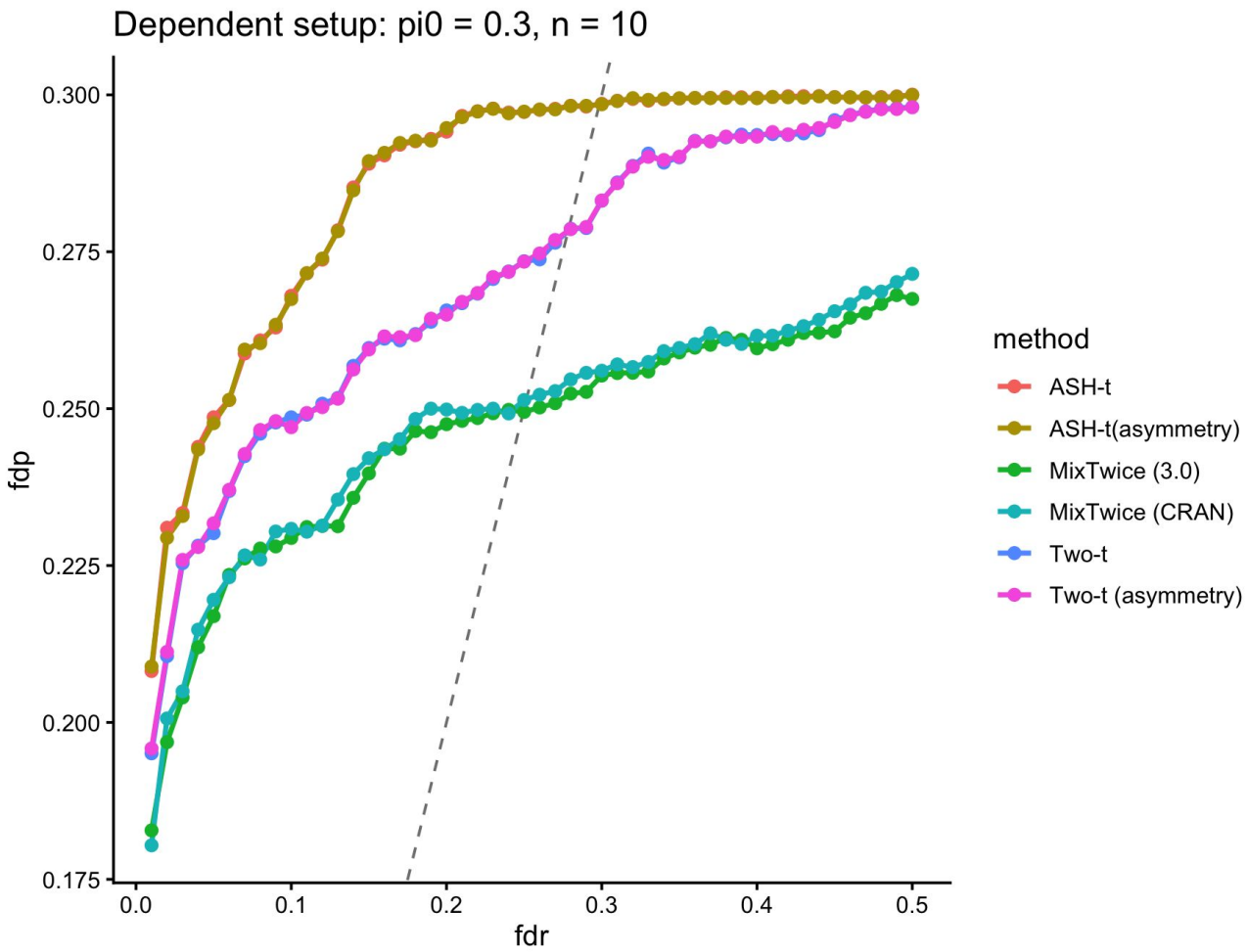
30% null, n= 10

Fdp: false calls/total calls

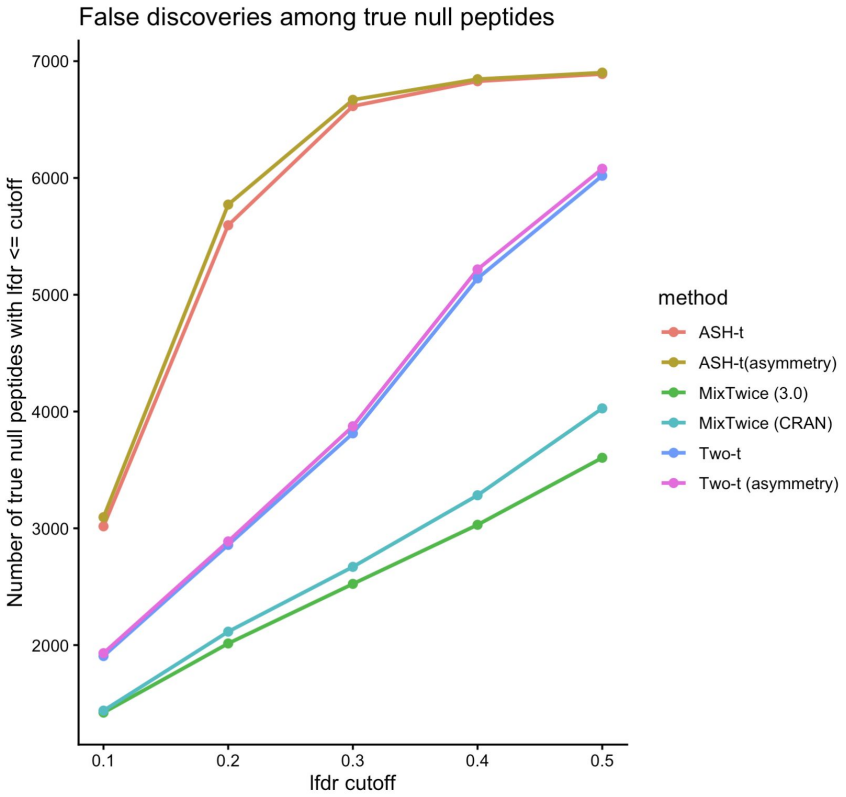







Dependent setup: pi0 = 0.3, n = 10
0.300
0.275
0.250
0.225
0.200
0.175
fdp
0.0
0.1
0.2
0.3
0.4
0.5
fdr
method
ASH-t
ASH-t(asymmetry)
MixTwice (3.0)
MixTwice (CRAN)
Two-t
Two-t (asymmetry)

70% null, n= 10


False discoveries among true null peptides
Number of true null peptides with lfdr <= cutoff
7000
6000
5000
4000
3000
2000
0.1
0.2
0.3
0.4
0.5
lfdr cutoff
method
ASH-t
ASH-t(asymmetry)
MixTwice (3.0)
MixTwice (CRAN)
Two-t
Two-t (asymmetry)

1. Case 1 L = 10000

$$\sigma_i^2 \sim h(\sigma^2), \qquad \sigma_i^2 \sim 0.88 \cdot 0.11\,\mathrm{Beta}(0.85, 6.5) \;+\; 0.12 \cdot 0.11\,\mathrm{Beta}(2.0, 4.2)$$

$$\pi_0 = 0.8 \quad (80\%\ \text{null},\ 20\%\ \text{non-null}),$$

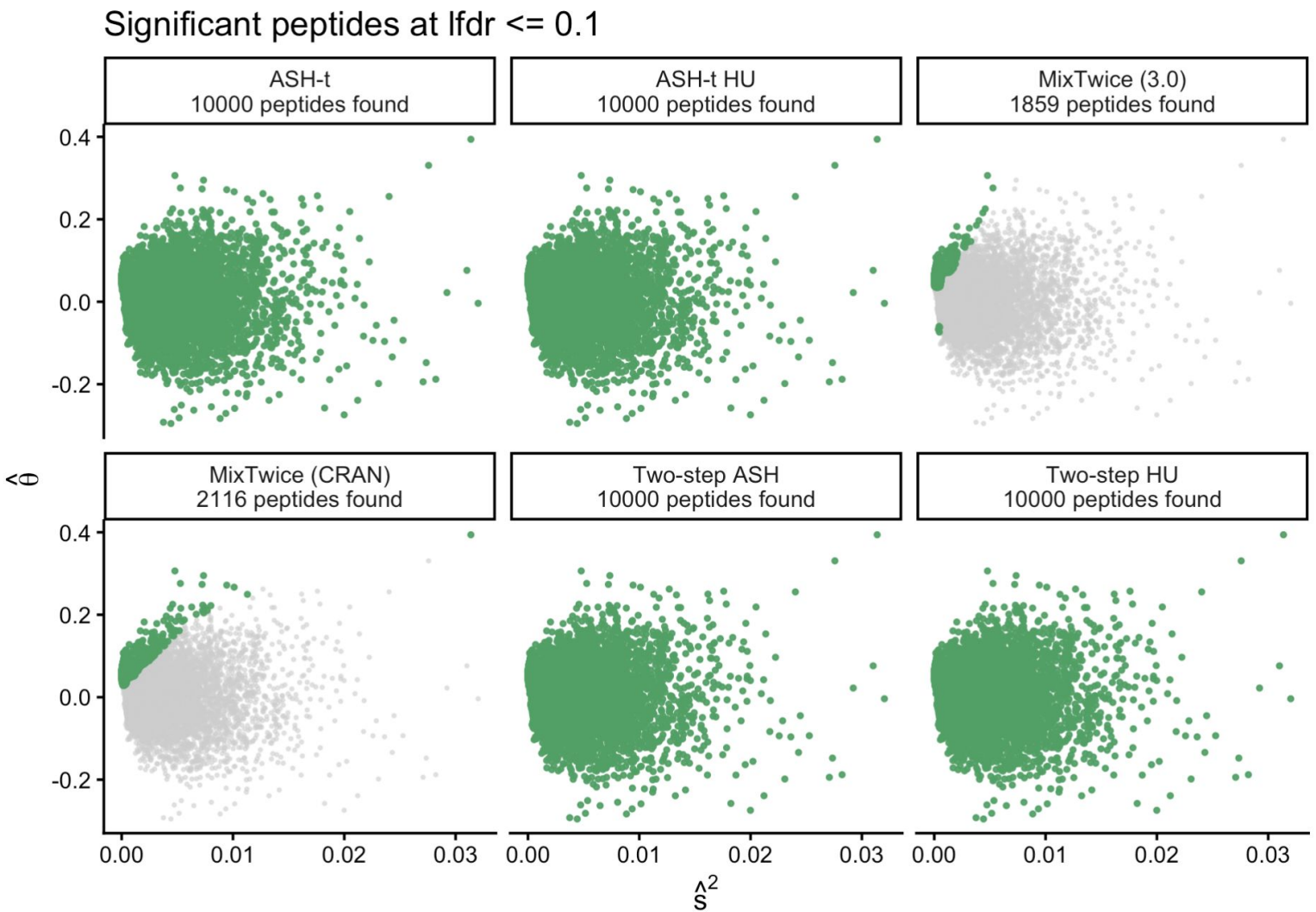
$$\gamma_i = \begin{cases} 1, & r_i^2 \le q_{0.2}(r^2), \\ 0, & r_i^2 > q_{0.2}(r^2), \end{cases}$$

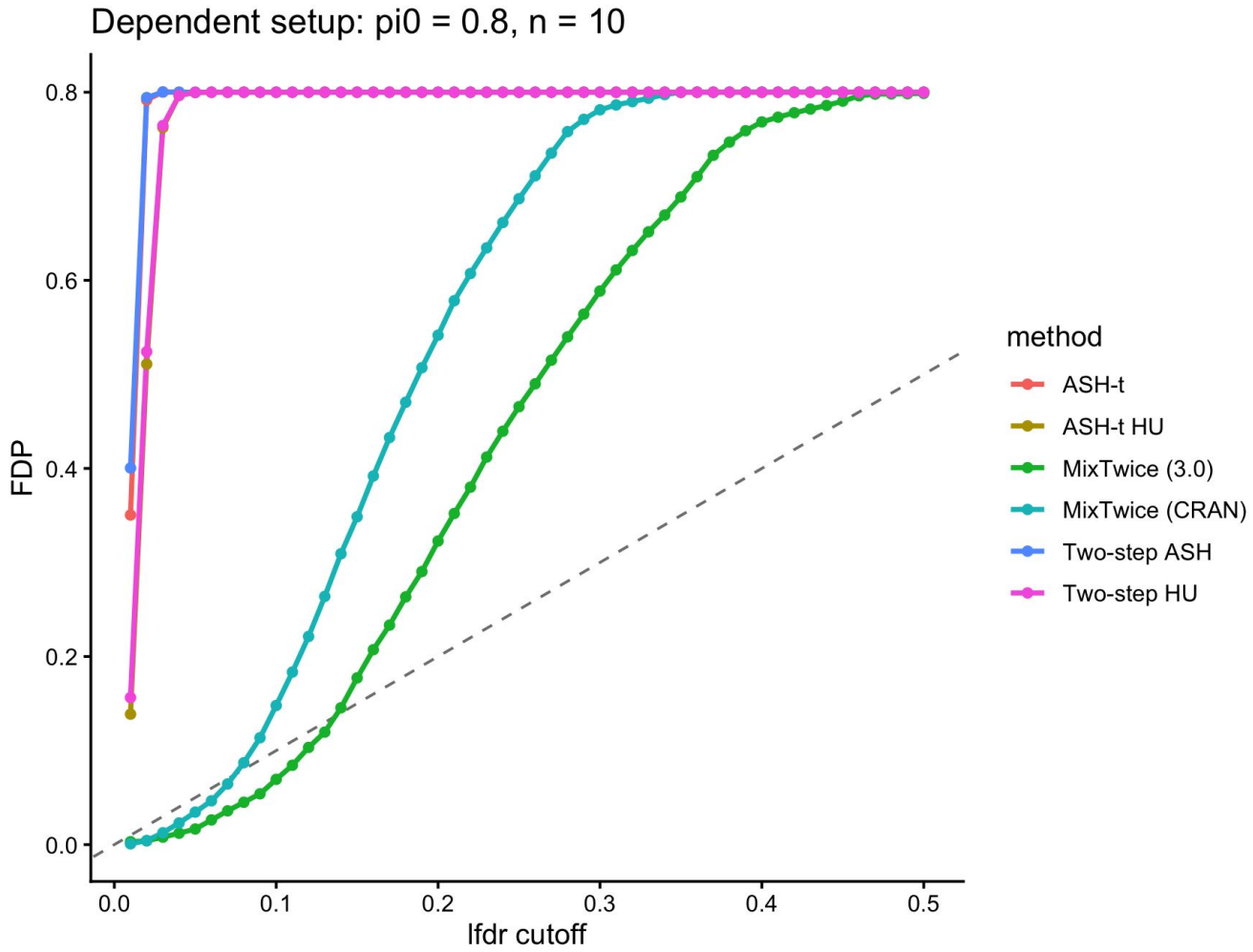
Mechanism:
Ash: g(theata) for all i
Data g(theta|ri^2) concentrated around low-variance

$$\theta_i \mid r_i^2, \gamma_i \sim \begin{cases} 0, & \gamma_i = 0, \\ N(0.055,\ 0.10\, r_i^2), & \gamma_i = 1. \end{cases}$$

$$x_i \mid \theta_i, r_i^2 \sim N(\theta_i, r_i^2),$$

$$s_i^2 \mid r_i^2 \sim \frac{r_i^2}{\nu}\chi_\nu^2, \qquad \nu = 2(n-1).$$


Significant peptides at lfdr <= 0.1
ASH-t
10000 peptides found
ASH-t HU
10000 peptides found
MixTwice (3.0)
1859 peptides found
MixTwice (CRAN)
2116 peptides found
Two-step ASH
10000 peptides found
Two-step HU
10000 peptides found
0.4
0.2
0.0
-0.2
0.00
0.01
0.02
0.03
$\hat{\theta}$
$\hat{s}^2$


Dependent setup: pi0 = 0.8, n = 10
FDP
0.0
0.2
0.4
0.6
0.8
0.0
0.1
0.2
0.3
0.4
0.5
lfdr cutoff
method
ASH-t
ASH-t HU
MixTwice (3.0)
MixTwice (CRAN)
Two-step ASH
Two-step HU

2. Case 2

$$l = 150000, \qquad n = 10, \qquad \nu = 2n - 2.$$

$$\gamma_i = \begin{cases} 0, & i < 100001, \\ 1, & i \geq 100001. \end{cases}$$

$$\sigma_i^2 \mid \gamma_i \sim \begin{cases} 0.08\,\mathrm{Beta}(0.9, 6.5), & \gamma_i = 0, \\ 0.08\,\mathrm{Beta}(1.2, 5.8), & \gamma_i = 1. \end{cases}$$

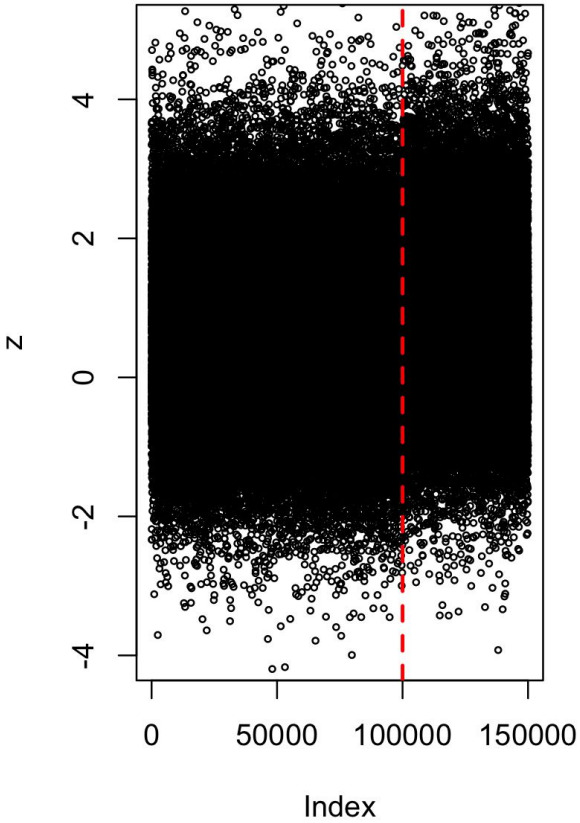
$$r_i^2 = \frac{2\sigma_i^2}{n}.$$

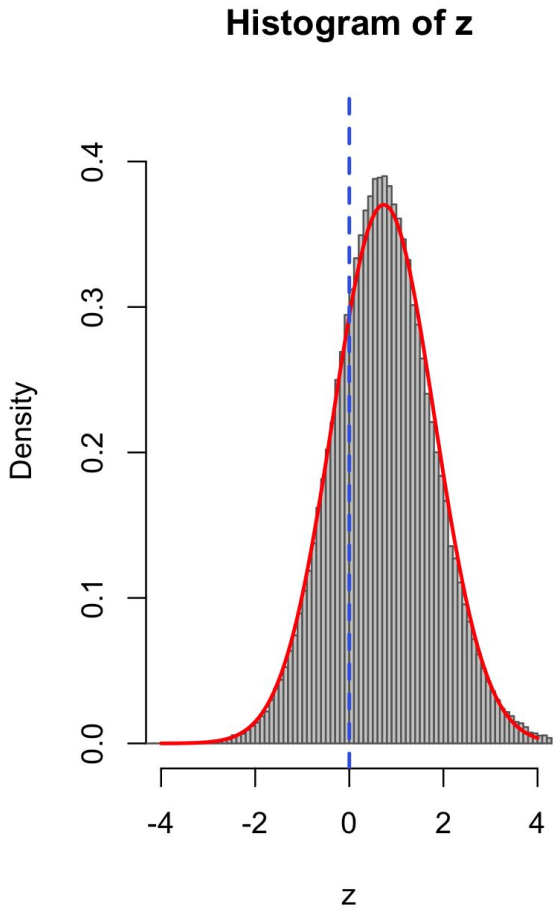
$$\theta_i \mid r_i^2, \gamma_i \sim \begin{cases} N(0.60\sqrt{r_i^2},\, 0.03^2 r_i^2), & \gamma_i = 0, \\ N(0.90\sqrt{r_i^2},\, 0.04^2 r_i^2), & \gamma_i = 1. \end{cases}$$

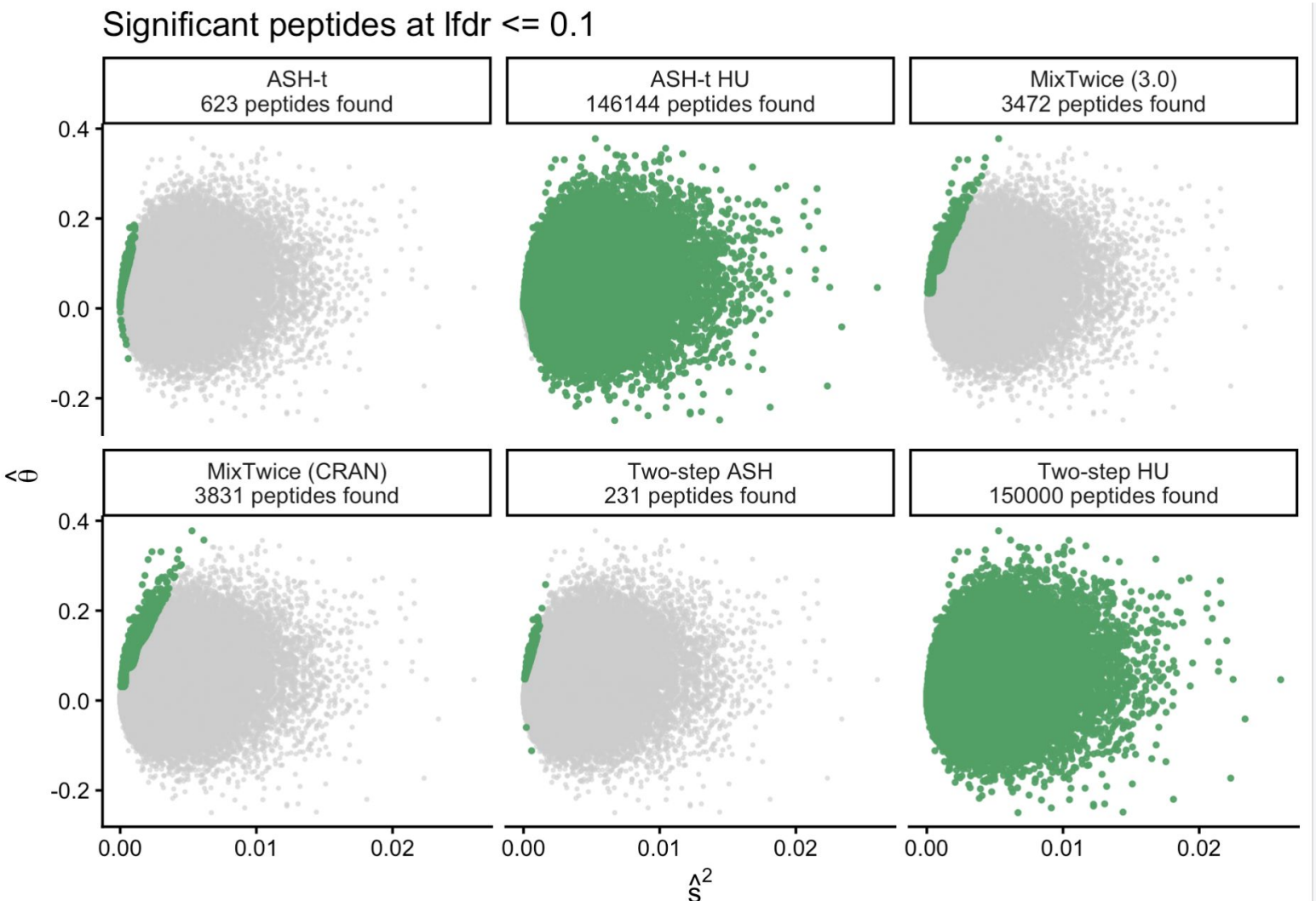
$$x_i \mid \theta_i, r_i^2 \sim N(\theta_i, r_i^2),$$

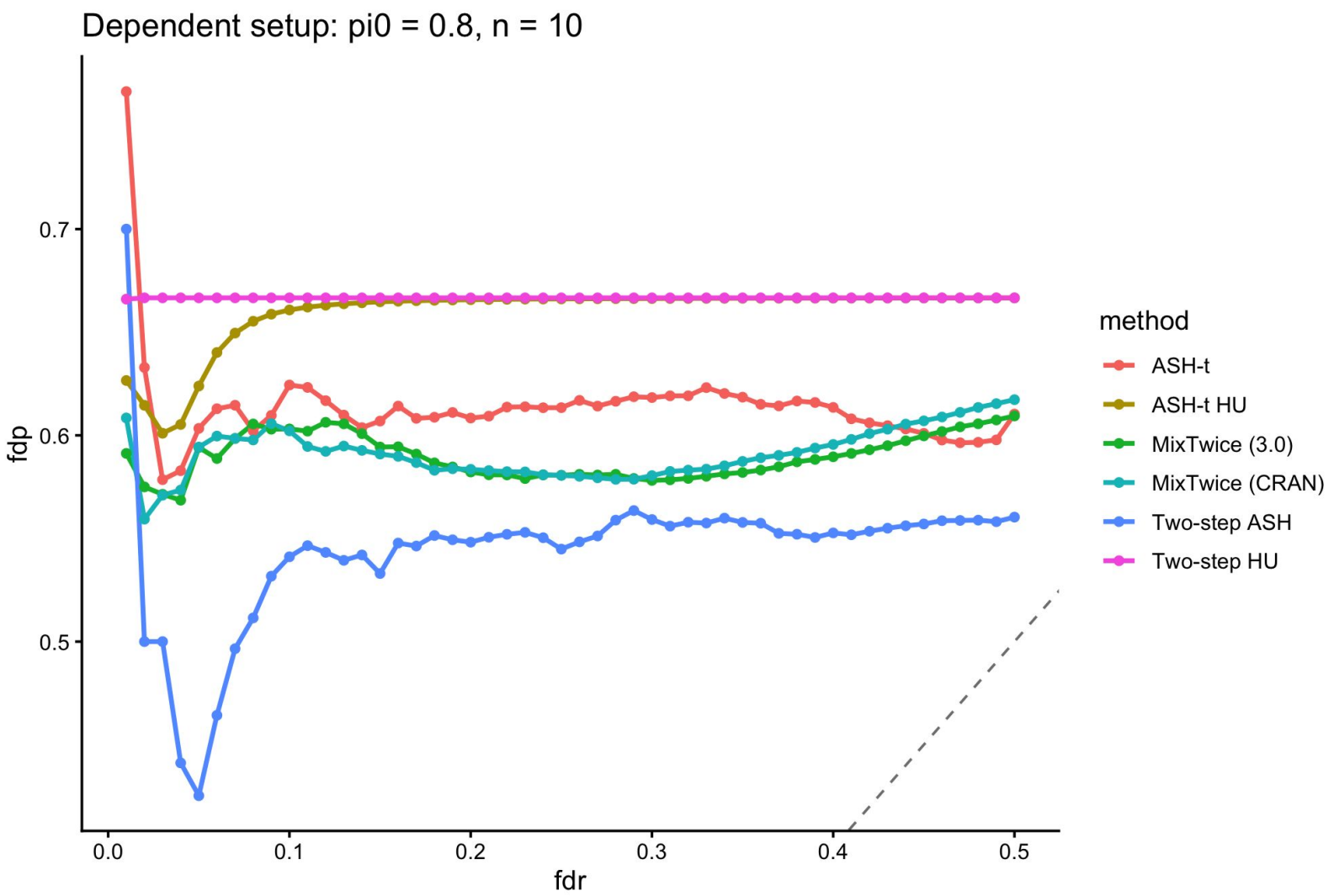
$$s_i^2 \mid r_i^2 \sim \frac{r_i^2}{\nu}\chi_\nu^2.$$

Mechanism:
Ash-Hu tries to explain the shift by putting almost all prior mass on positive effects


z
4
2
0
-2
-4
0
50000
100000
150000
Index


Histogram of z
Density
0.0
0.1
0.2
0.3
0.4
-4
-2
0
2
4
z


Significant peptides at lfdr <= 0.1
ASH-t
623 peptides found
ASH-t HU
146144 peptides found
MixTwice (3.0)
3472 peptides found
MixTwice (CRAN)
3831 peptides found
Two-step ASH
231 peptides found
Two-step HU
150000 peptides found
0.4
0.2
0.0
-0.2
0.00
0.01
0.02
$\hat{\theta}$
$\hat{s}^2$


Dependent setup: pi0 = 0.8, n = 10
fdp
0.7
0.6
0.5
0.0
0.1
0.2
0.3
0.4
0.5
fdr
method
ASH-t
ASH-t HU
MixTwice (3.0)
MixTwice (CRAN)
Two-step ASH
Two-step HU